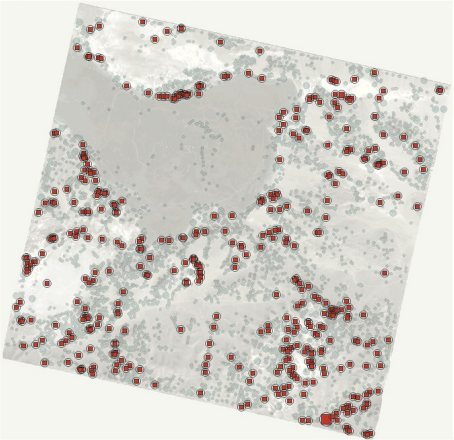
ALLOTROPE21 LABS

# Allotrope

## Anomaly detection, material identification and atmospheric correction in hyperspectral satellite imagery

Indradhanu, our own spectral foundation model, learns ordinary ground; what it cannot rebuild is flagged, then named man-made or natural with its closest material class.

AIGC Stage-III evaluation · PS-11
4 November 2026, 09:00



PRISMA, 14 Feb 2025: 405 of 5,544 anomaly objects called man-made.

| 0.189 | 449 / 909 | 86–94 % |
|---|---|---|
| pAUC, false alarms ≤1 % | objects in the top 1 % | man-made targets found |
| previous release 0.170 | no model 377 | at quarter-pixel fill; ACE 66–77 % |

# Problem and data

### THE TASK

Find small, unusual objects in hyperspectral images with no target list given in advance, and say what material each is made of. At Stage II our model (700+ scenes) raised false alarms on urban scenes, land–water edges, cloud contours and snow; Stage III widened the data and moved the model's attention to the spectrum.

### DATA PROCURED FOR STAGE III

| | |
|---|---|
| pretraining | SpectralEarth, all of it: ≈3.3 TB, ≈11,636 EnMAP scenes, 539,755 patches |
| curated | 1,300+ EnMAP, PRISMA and AVIRIS-NG scenes; India and borders; urban, forest, desert, hills, coast; 2–40 % cloud |
| L1 / L2 pairs | 60 EnMAP and 30 PRISMA (L1, L2C, L2D) for atmospheric correction; 50+ public Tanager scenes |

### SENSORS

| sensor | bands in the file | pixel | role |
|---|---|---|---|
| PRISMA | 239 (185 used) | 30 m | test bench; fine-tuning; namer |
| EnMAP | 224 | 30 m | pretraining; fine-tuning; namer |
| AVIRIS-NG | 372 | varies | fine-tuning; namer; field targets |
| Tanager | 426 (368 good) | 30 m | end to end; never trained on |

### THE TEST BENCH

Four labelled PRISMA L2D scenes, **909 ground-truth objects** in three tiers (easy, medium, subtle). Every method is scored on the same valid pixels and the same object maps.

### MEASURES

| measure | what it means | chance |
|---|---|---|
| **pAUC** | area under the detection curve while false alarms stay ≤1 % of background, scaled so 1 is perfect | ≈0.005 |
| top 1 % | object found if any pixel lies within 1 px of the top 1 % of scores | 10.2 % (93) |
| top 0.1 % | the same, ten times stricter | 1.1 % |
| ROC-AUC | ranking over all thresholds | 0.5 |
| bootstrap | resample the 909 objects 2,000 times; 95 % interval of the difference | |

Bench resolution: about ±0.005–0.01 pAUC.

# Pipeline

**Band filter**
native bands; dead bands out; no-data filled

→

**Segmentation**
V2 head: land, water, cloud, shadow, snow

→

**Scoring**
V2 rebuilds each pixel hidden; Mahalanobis score

→

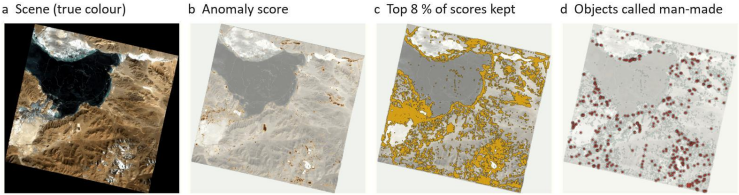
**Threshold**
analyst keeps the top scores; boolean mask

→

**Spectral match**
namer: man-made or natural, class; SAM, ACE

→

**Export**
GeoTIFF mask, shapefiles with material attributes



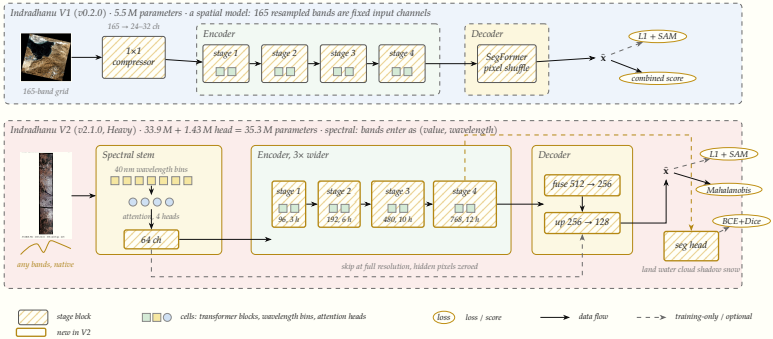

One PRISMA scene (14 Feb 2025, 1237×1202 px) through the chain: 30.3 % no-data; thick cloud (4.9 %) left out; the analyst keeps the top 8 %, 118,950 pixels; 5,544 objects named, 405 man-made (red).

TIME PER PRISMA SCENE

| | |
|---|---|
| V2 Heavy, 16-thread CPU | 454 s |
| previous release | 274 s |
| residual Mahalanobis | 3–4 s |
| spectral namer | 8–9 s |

The matcher's natural call removes natural anomalies after scoring, with a tunable confidence.

# Spatial to spectral




## V1, Stage II

A spatial SegFormer MAE on one resampled 165-band grid; a 1×1 compressor ties weights to band position: it knows "input 87", not 1,650 nm.

## V2, Stage III

A wavelength-aware stem turns each pixel into tokens over 40 nm bins and attends over them: PRISMA's 185, EnMAP's 224, AVIRIS-NG's 372 bands read natively.

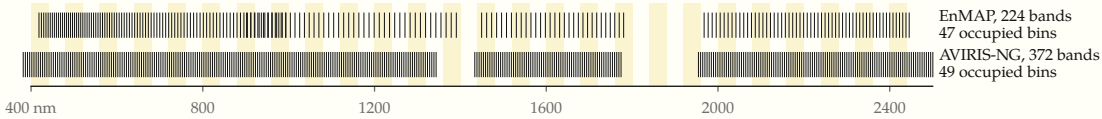
## What made it a foundation model

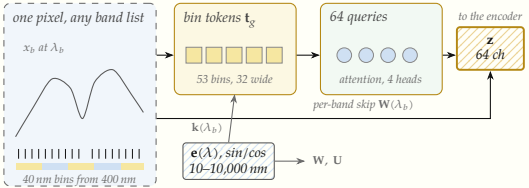
SpectralEarth pretraining (512,849 patches), band-subset and wavelength-jitter augmentation, 3× wider encoder, progressive decoder, segmentation head.

*pAUC 0.165 (V1) → 0.170 (V2.0) → 0.189 (V2 Heavy) on the 909-object bench, thick cloud excluded.*

# Spectral stem

TWO REAL BAND LISTS, ONE TOKEN GRID: EVERY SENSOR'S BANDS FALL INTO THE SAME 40 NM BINS




THE MECHANISM, PER PIXEL

$$\mathbf{e}(\lambda) = [\sin \tfrac{2\pi\lambda}{P_i}, \cos \tfrac{2\pi\lambda}{P_i}]_{i=1}^{16}, \qquad P_i \text{ from 10 to } 10^4 \text{ nm}$$

$$\mathbf{t}_g = \langle x_b\, \mathbf{k}(\mathbf{e}(\lambda_b))\rangle_{b\in g} + \beta(\mathbf{e}(\bar{\lambda}_g)), \qquad g = 1,\dots,53 \text{ (40 nm)}$$

$$z_j = \mathrm{Attn}(\mathbf{q}_j;\, \{\mathbf{t}_g\}) + \langle x_b\, W_j(\mathbf{e}(\lambda_b))\rangle_{\text{all } b}, \qquad j = 1,\dots,64$$

$$\hat{x}(\lambda) = \mathbf{U}(\mathbf{e}(\lambda)) \cdot \hat{\mathbf{z}} + c(\mathbf{e}(\lambda)) \qquad \text{at any requested } \lambda$$

$$\langle f\rangle_B = \textstyle\sum_{b\in B} w_b f_b \,/\, \sum_{b\in B} w_b, \qquad w_b = \text{nm band } b \text{ covers}$$

$x_b$: band $b$'s reflectance, normalised by one fixed curve over wavelength (not per scene). $\mathbf{k}, \mathbf{W}, \mathbf{U}, c$: small networks of $\mathbf{e}$, so no weight is tied to a band position. 64 learned queries attend over the occupied bins with 4 heads. A token *is* a wavelength range: a band at 1,655 nm reuses what was learned at 1,650 nm.

- **Band-list augmentation**: 80 % of batches see a random subset (≥40 bands), wavelengths shifted up to 3 nm. Before it, native PRISMA: ROC-AUC 0.205, resampled 0.725.
- 17,825 parameters; ≈6× faster than attention over every band.

# Training

TWO STAGES ON ONE RENTED H100 (80 GB)

| stage | data | model | epochs | settings, time |
|---|---|---|---|---|
| pretraining | SpectralEarth (EnMAP), 202 bands; 512,849 training patches, 30,000 per epoch | 33.9 M | 80 | lr $3\times10^{-4}$, batch 32, bf16; ≈9 h |
| fine-tuning | EnMAP 210 scenes (14,542 tiles), PRISMA 223 (14,802), AVIRIS-NG 79 lines (22,377); mix 40/35/25 % | 35.3 M (+1.43 M head) | 10+8+6 | lr $10^{-4}\rightarrow3\times10^{-5}$; ≈5.5 h |
| selection | four labelled PRISMA scenes | 35.3 M | ep. 5 of 6 | epochs 4 / 6: 0.184 / 0.187 |

OBJECTIVE, ON HIDDEN PIXELS $M$ (65 % OF EACH PATCH)

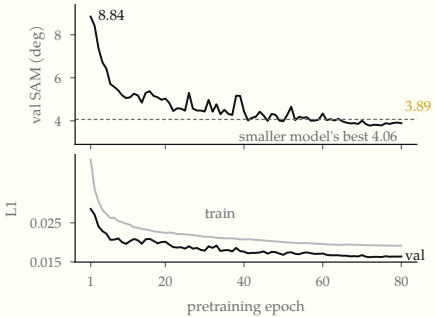
$$\mathcal{L} = \underbrace{\frac{1}{C|M|}\sum_{p\in M}\sum_{c}|\hat{x}_{c,p}-x_{c,p}|}_{L_1} + \lambda(t)\ \underbrace{\frac{1}{|M|}\sum_{p\in M}\angle(\mathbf{x}_p,\hat{\mathbf{x}}_p)}_{\text{spectral angle}} + 0.1\,\mathcal{L}_{\text{seg}}$$

$\lambda(t)$ rises 0 → 0.5 over the first ten epochs; $\mathcal{L}_{\text{seg}}$ = BCE + Dice, fine-tuning only, shadow and snow ×3.

FEEDING THE GPU

150-patch archives (≈1 GB, 3,419 for training) read straight from storage gave 18 patches/s; 200 at a time are copied to local disk and swapped every five epochs: 81 patches/s, 18.4 GB peak, 3,200 archives used. Scenes within 30 km of the bench are kept out.

PRETRAINING, VALIDATION

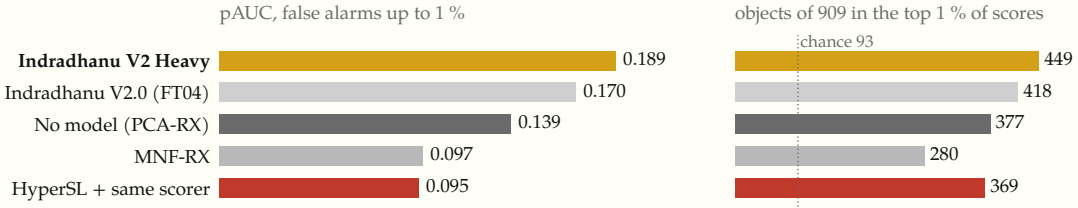



*Validation spectral angle 8.84° → 3.89°, below the smaller model's best 4.06°; epoch 72 seeds fine-tuning.*

Epoch 5 was picked on the bench, so its bench score is slightly optimistic; its neighbours sit within 0.005.

# Detection results

909 labelled objects, four PRISMA scenes, thick cloud excluded; water stays in (vessels are targets)




## The score: residual Mahalanobis

$$\mathbf{r}_p = \mathbf{x}_p - \hat{\mathbf{x}}_p, \qquad \mathbf{z}_p = \Lambda_d^{-1/2}\mathbf{U}_d^{\top}(\mathbf{r}_p - \mu_r),\ d = 30$$

$$s(p) = \left\| \mathbf{z}_p - \tfrac{1}{|\mathcal{R}_p|} \textstyle\sum_{q \in \mathcal{R}_p} \mathbf{z}_q \right\|^2$$

Whiten the residual in its 30 strongest directions; compare with a ring between 21×21 and 61×61. Direction, not size: mean ROC-AUC **0.662 → 0.854**, same model.

## Paired bootstrap against FT04

| | |
|---|---|
| epoch 5 | **+0.0196** [+0.0072, +0.0316] |
| epoch 4 | +0.0148 [+0.0029, +0.0259] |
| epoch 6 | +0.0179 [+0.0055, +0.0292] |
| no model − V2 | −0.047 [−0.060, −0.033] |

Top 1 % by tier: easy 58.2 → **62.2**, medium 39.3 → **43.9**, subtle 40.3 → **42.0** %. At top 0.1 % the two tie (+0.1 pts).

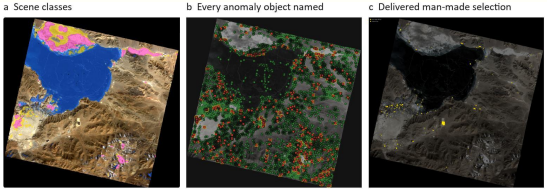
## Per scene, V2 Heavy

| | pAUC | ROC-AUC |
|---|---|---|
| Set-1 | 0.2617 | 0.936 |
| Set-2 | 0.1874 | 0.874 |
| Set-3 | 0.0861 | 0.781 |
| Set-4 | 0.2217 | 0.888 |

The delivered chain reproduces all four to four decimals. Leaving out thick cloud only: 0.175 → 0.189.

Public benchmark HAD100: appendix A6.

# Detection examples




PRISMA. (a) V2 classes: water blue, cloud pink (left out), haze yellow. (b) 5,544 objects, man-made red, natural green. (c) A delivered selection.

### THE SAME CHAIN ON THREE SENSORS

| | Tanager | PRISMA | AVIRIS-NG |
|---|---|---|---|
| anomaly objects | 1,103 | 5,544 | 2,375 |
| man-made, $P > 0.5$ | 272 | 405 | 1,290 |
| man-made, $P > 0.25$ | 585 | 2,263 | 1,716 |
| oil/fuel or plastic | 247 | 141 | 936 |

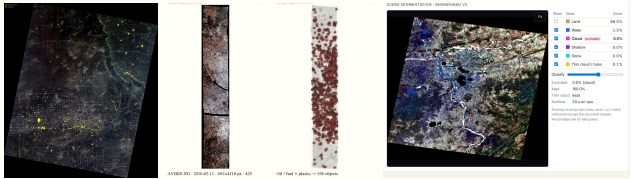
Tanager was never in training; GHOSt (Golmud, 8 m) ran end to end on evaluation day.

### THE 405 MAN-MADE OBJECTS

| | | | |
|---|---|---|---|
| concrete/asphalt | 244 | metal | 16 |
| oil/fuel | 116 | textile | 3 |
| plastic | 25 | wood/paper | 1 |

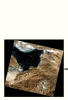
### SEGMENTATION HEAD, VALIDATION IoU

water 0.825, cloud 0.749. Its thin-cloud class also fires on haze (75.4 % of Set-4, 209 of 230 targets), so only thick cloud is left out.



Tanager, man-made | AVIRIS-NG | oil + plastic | EnMAP, classes

# Atmospheric correction



*L1 radiance or TOA, any file*

**Radiometric**

**Read** native or generic; level detected, not assumed

**To radiance** gain, offset, scale per sensor

**Atmospheric (ISOFIT 4.1.5)**

**Look-up table** 6S + sRTMnet; AOT, $H_2O$, elevation, geometry

**Optimal estimation** on superpixels; surface prior

**Every pixel** analytical or empirical line, GPU

**Surface**

**Post** outliers, terrain, adjacency, snow

**Harmonise** towards the vendor L2, per sensor

**L2 surface reflectance** GeoTIFF + BSQ, wavelengths

*Copernicus DEM*

*vendor L2 (fit only)*

## RADIANCE MODEL AND OPTIMAL ESTIMATION

$$\rho_{\mathrm{TOA}} = \frac{\pi d^2 L}{E_0 \cos\theta_s} = \rho_{\mathrm{path}} + \frac{T^{\downarrow} T^{\uparrow} \rho_s}{1 - S\rho_s}$$

$$\hat{\mathbf{x}} = \arg\min_{\mathbf{x}} \; \|\mathbf{y} - F(\mathbf{x})\|^2_{\mathbf{S}_\epsilon^{-1}} + \|\mathbf{x} - \mathbf{x}_a\|^2_{\mathbf{S}_a^{-1}}$$

State $\mathbf{x}$: surface reflectance $\rho_s(\lambda)$, aerosol optical thickness, water vapour. $F$: the look-up-table forward model; $\mathbf{x}_a$, $\mathbf{S}_a$: surface prior; $\mathbf{S}_\epsilon$: instrument noise. Adjacency: $\rho = \rho_{\mathrm{ret}} + q(\rho_{\mathrm{ret}} - \rho_{\mathrm{bg}})$, 1 km kernel. Inverts within 350–1360, 1410–1800, 1970–2500 nm. Superpixels of 50 for satellites; 400 with an empirical line for 4–16 Mpix AVIRIS-NG lines.

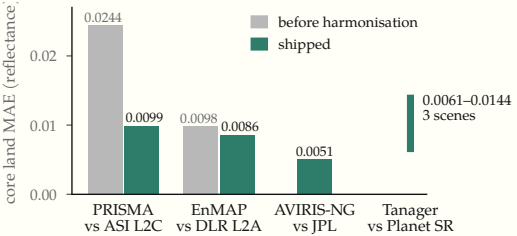
## PER-SENSOR RULES

| sensor | to radiance | aerosol | terrain, adjacency | bands kept |
|---|---|---|---|---|
| PRISMA | DN / scale − offset; VNIR/SWIR split 950 nm | NASA POWER climatology ±0.15 | off | 206 |
| EnMAP | DN × gain + offset | free, seasonal limits | on | 218 |
| AVIRIS-NG | units only | 0.10–0.14, as JPL | off | 380 |
| Tanager | units only | free, seasonal limits | on | 382 |

Terrain only where the vendor does it: SWIR $r$ 0.947 → 0.980 on mountains. Snow by NDSI (precision 0.91–1.00). Reflectance > 2.5 kept and flagged: flares and fires are anomalies.

# Correction accuracy

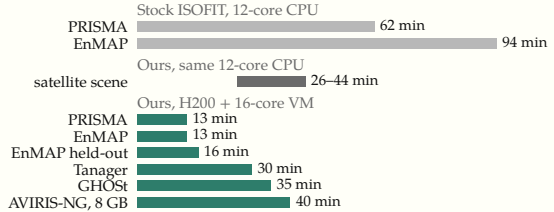
HELD-OUT SCENES AGAINST THE VENDOR L2, 460–2450 NM CORE




Two good L2 products differ by 0.01–0.02. A per-sensor model of the retrieved atmosphere (AOT, $H_2O$, sun zenith, elevation) maps ours to the vendor's; fitted on 15–27 scenes, scored once on 5 held-out. PRISMA: all five improved. EnMAP: blue bands only, 460 nm bias +0.023 → +0.001.

EnMAP's blue range was chosen after the held-out scores were seen. Aim: one reflectance domain, so one anomaly model serves every sensor.

RUN TIME PER SCENE




HOW

| | |
|---|---|
| exact | 6S aerosol-only call removed (1.9×), scattering cached over water vapour (5×); GPU per-pixel solve 3.2×, difference $\leq 6\times10^{-8}$ |
| measured | EnMAP prior on true band centres 53 → 30 min; AVIRIS-NG superpixels 400: window 40.8 → 7.5 min, error 0.0056 → 0.0057; full 4 Mpix line 373 → 41 min |
| robust | input level detected by $O_2$/$H_2O$ absorption tests, 28/28 right; ambiguity stops the run and names the fix |

# Material identification

TWO MATCHERS ON EVERY DETECTED PIXEL

USGS splib07 + ECOSTRESS, resampled to the sensor: seven man-made classes (oil/fuel, plastic, textile, wood/paper, paint, metal, concrete/asphalt) and a natural shelf; 378 / 338 / 380 spectra (EnMAP / PRISMA / AVIRIS-NG).

**SAM**: smallest angle $\theta_j = \arccos \frac{\mathbf{x}\cdot\mathbf{s}_j}{\|\mathbf{x}\|\|\mathbf{s}_j\|}$; no training, but blind to the ground filling the rest of the pixel.

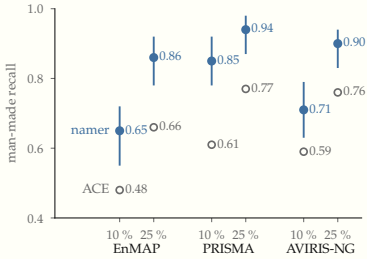
**Spectral namer** (347,685 parameters) on **V2's spectral stem**: one tower reads the pixel and its ring mean (3–7 px away), the other the library.

$$s_j = \cos(\mathbf{z}, \mathbf{e}_j)/\tau, \quad S_c = \log \sum_{j \in c} e^{s_j}$$

$$P(\text{man-made}) = \sigma\left( \mathrm{LSE}_{c \neq \text{nat}}\, S_c - S_{\text{nat}} + b \right)$$

Level 1: share of the match on man-made spectra; level 2: best class. Trained on library spectra planted at 5–100 % fill in real EnMAP, PRISMA, AVIRIS-NG pixels, 20 % of each class held out. New spectra need no retraining.

LEVEL 1 AT ACE'S FALSE-ALARM RATE




*Quarter-pixel fill: 86–94 % of man-made targets found; ACE 66–77 %.*

LEVEL 2, CLASS CORRECT

| fill 10 % | namer | ACE |
|---|---|---|
| EnMAP | **0.59** [0.48, 0.70] | 0.24 |
| PRISMA | **0.69** [0.53, 0.81] | 0.41 |
| AVIRIS-NG | **0.65** [0.53, 0.76] | 0.38 |
| fill 25 % | | |
| EnMAP | **0.68** [0.55, 0.79] | 0.47 |
| PRISMA | 0.70 [0.54, 0.82] | 0.60 |
| AVIRIS-NG | 0.70 [0.57, 0.82] | 0.58 |

Planted sub-pixel targets in PRISMA scenes, 10 % fill, 8 classes: namer **0.53**; best training-free 0.44; HyperSL-based 0.27–0.30.

# Field targets

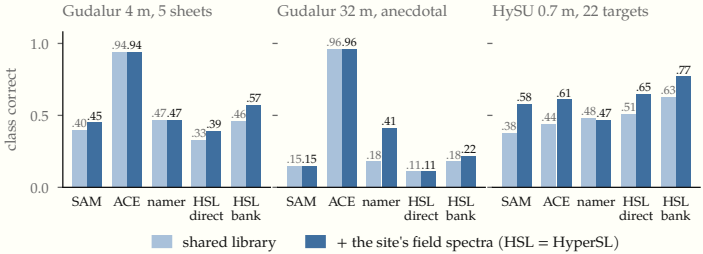
REAL TARGETS LAID OUT IN THE FIELD

**HySU** (DLR, airborne, 0.7 m): bitumen, red-painted metal, blue, red and green synthetic fabric, squares 0.25–3 m; 135 bands, 417–903 nm, a range the namer never trained on. 22 targets, 236 target pixels, 3,000 background.

**Gudalur** (AVIRIS-NG, Nilgiris 2018, 4 m): five 16×16 m sheets (green, red, yellow, black nylon; white cotton) with field spectra; 80 pixels, 353 bands. Also coarsened to 32 m at four offsets: each sheet ≤ a quarter pixel, 27 pixels (anecdotal).

Namer here has textile held out, so textile is unseen; nylon may count as plastic or textile, bitumen as concrete or oil.

LEVEL 2: CLASS CORRECT, SHARED LIBRARY AND WITH THE SITE'S OWN SPECTRA




| level 1: AUC of the man-made call | Gudalur 4 m | Gudalur 32 m | HySU |
|---|---|---|---|
| SAM | 0.69 / 0.70 | 0.39 / 0.39 | 0.65 / 0.73 |
| ACE | 0.58 / **0.89** | 0.58 / **0.86** | 0.63 / **0.81** |
| spectral namer | 0.15 / 0.15 | 0.17 / 0.18 | 0.58 / 0.61 |
| HyperSL, mixture bank | 0.51 / 0.59 | 0.15 / 0.19 | 0.57 / 0.80 |

ACE names the Gudalur sheets best (5 of 5 at 4 m, 18 of 19 at 32 m). The namer's man-made call is *inverted* on the sheets: a 16 m sheet is not a pixel unlike its ring. On HySU the HyperSL mixture bank names the most (16 of 22; 18 with field spectra). Small sets: examples, not rates.

# Against HyperSL

A PUBLIC SPECTRAL FOUNDATION MODEL, RUN UNMODIFIED

Per-pixel masked autoencoder, 9.2 M parameters, EnMAP and DESIS reflectance; hides 95 % of a pixel's bands and rebuilds them from the same pixel.

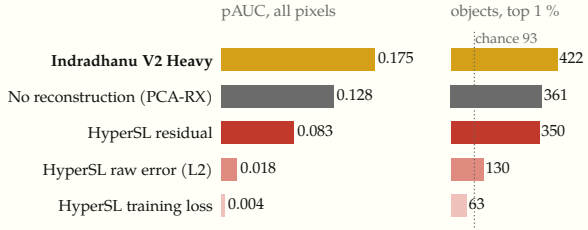
**1 Self-reconstruction.** An unusual pixel rebuilds as well as an ordinary one: its own loss as a score finds 63 objects, fewer than random (93). Indradhanu hides whole pixels.

**2 Per-scene min-max.** $x'_b = (x_b - m_b)/(M_b - m_b)$ from the scene: $(0.20, 0.50)$ becomes $(0.5, 1.0)$ in one scene, $(0.43, 0.75)$ in another; an angle moves $15.1° \rightarrow 36.9°$ with no change in reflectance. A library spectrum has no scene: 41 % of ECOSTRESS values clipped.

**3 Band order, not wavelength.** List neighbours straddle water-vapour gaps and differ per sensor. Indradhanu bins by wavelength.

**4 Cloud-free pretraining.** Cirrus recall 28.9 % from its features against 67.9 % from raw bands.

SAME BENCH, SAME SCORER, ALL PIXELS




| paired difference [95 %] | pAUC | objects (pts) |
|---|---|---|
| HyperSL − no model | −0.045 [−0.059, −0.032] | −1.2 [−4.0, +1.5] |
| HyperSL − V2 Heavy | −0.092 [−0.106, −0.078] | −7.9 [−10.8, −5.2] |

| unmixing RMSE, true endmembers | DC1 / DC2 / DC3 | EGU | Jasper | Urban |
|---|---|---|---|---|
| reflectance, best physical | 0.000 / 0.000 / 0.000 | 0.018 | 0.050 | 0.059 |
| HyperSL internal space | 0.253 / 0.113 / 0.261 | 0.142 | 0.106 | 0.170 |

*Not adopted: it lowers detection below no model, bends mixing (0.4–0.7 of the path length), and its matching trails plain reflectance.*

# Deliverables and limits

## Per scene and selection

- Boolean anomaly mask, GeoTIFF, 0/1 on the scene's own UTM grid; **no heatmap**.
- Pixel and object shapefiles and a table, latitude and longitude.
- Every pixel and object: namer call, class and confidence, ACE family match, best SAM match and its angle.
- L2 surface reflectance from L1, GeoTIFF and BSQ with wavelengths.

## On the evaluation VM

| | scoring | L1 → L2 |
|---|---|---|
| time | <30 s | 16 min |
| CPU, 16 cores | ≈10 % | ≈90 % peak |
| RAM, 64 GB | ≈24 % | ≈10 % |
| GPU, H200 | ≈3 % | ≈4 % |

## Evaluation scene: GHOST, Golmud airport

GeoTIFF input; airstrip, a smaller field and buildings on bare soil; visible striping. Two scorings: Heavy + RX hybrid, and Heavy alone. Natural calls removed: ≈8,000 → ≈3,000 and ≈5,500 → ≈3,700 pixels submitted.

| man-made class | sub. 1 | sub. 2 | namer's match | SAM's match |
|---|---|---|---|---|
| plastic | 39.6 | 30.3 | fiberglass | cerium oxide |
| metal | 24.2 | 21.8 | coated steel girder | calcite |
| concrete/asphalt | 23.1 | 23.4 | construction concrete | lead white |
| oil/fuel | 7.7 | 16.8 | oiled marsh plant | buckbrush |
| textile | 2.2 | 3.7 | terrycloth fabric | grass |
| wood/paper | 2.2 | 3.7 | particle board | grass |
| paint | 1.1 | 0.3 | olive green paint | grass |

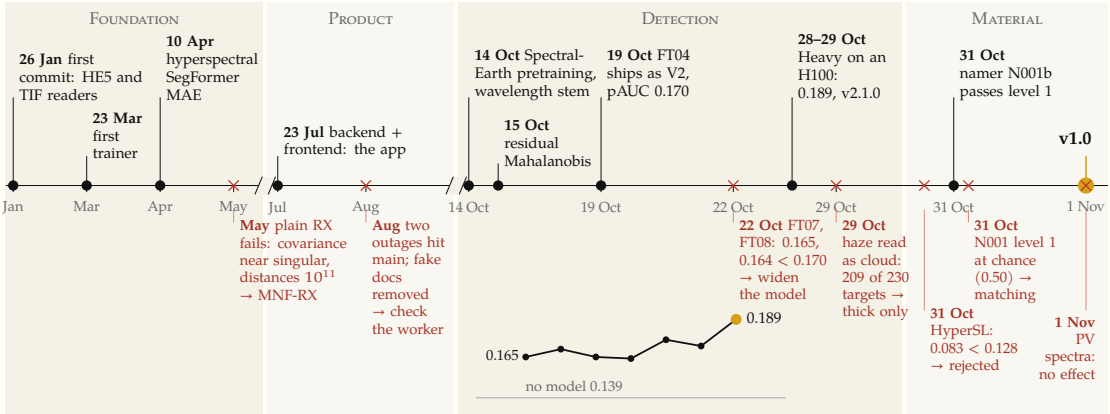
Share of man-made pixels, %. SAM's nearest spectra are minerals and plants.

## Limits

- **Large targets.** The namer compares a pixel with its ring; roads, roofs and big sheets defeat it. SAM and ACE guide there.
- **Open set.** Unseen textile: 46–47 % still called man-made, then a confident wrong class; abstention flags 1–4 %. In-scene solar panels changed nothing (0.248 → 0.248).
- **Small sets.** 5 sheets at Gudalur, 22 targets at HySU; four PRISMA scenes, ±0.005–0.01 pAUC.
- **Threshold** is an analyst choice; the evaluation scenes have no ground truth.
- **Cost.** Heavy 454 s per PRISMA scene on CPU, FT04 274 s.

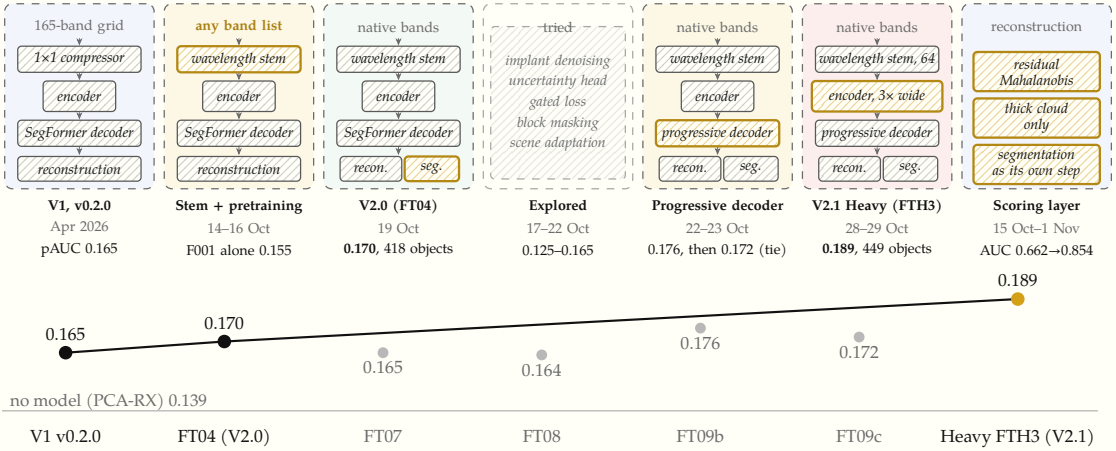
Next: field spectra for camouflage and coatings, a glass and solar-panel class, a labelled bench beyond PRISMA.

# Journey




From a spatial pipeline to a spectral foundation model: **any sensor in, materials out.**

# A1 Model generations




*Shipped generations in ink, experiments that did not beat the shipped model in grey; each column adds the part outlined in ochre.*

Also tried, not shipped: uncertainty head 0.125 / 0.163; per-scene adaptation 0.160. Pretrained only: 0.155 (FT04 size), 0.166 (Heavy).

# A2 Namer runs

| run | change | result |
|---|---|---|
| N001 | linear level-1 head | level 1 at chance, 0.49–0.52 (0.50 at full fill, where ACE gets 0.87) |
| N001b | level 1 by matching | passes on all three sensors, +0.20 / +0.17 / +0.14 over ACE; **shipped** |
| N002 | textile held out | open set: 46–47 % of textile still called man-made |
| N003 | mixed spectra in a class, warped wavelengths, dropout | matched N001b on EnMAP, no gain; N001b ships |
| PV | 50 in-scene solar-panel spectra | no effect, 0.248 → 0.248 |

N003 AGAINST THE SHIPPED N001B, ENMAP, EPOCH 40

| measure | N001b (shipped) | N003 |
|---|---|---|
| level 1 accuracy, fill 10 % | 0.679 | 0.677 |
| level 2 class accuracy | 0.587 | 0.588 |
| man-made recall at ACE's false-alarm rate | 0.647 | 0.639 |

PLANTED PIXELS FROM REAL TILES OF THREE SENSORS

$$\mathbf{x} = \mathbf{g} \odot (a\,\mathbf{t} + (1-a)\,\mathbf{b}) + \mathbf{n}, \quad a \sim \text{log-uniform}[0.05, 1]$$

**t** a library spectrum (50 % man-made, 25 % natural, 25 % none), **b** the real background, **g** a smooth gain within ±10 %, **n** the tile's own noise.

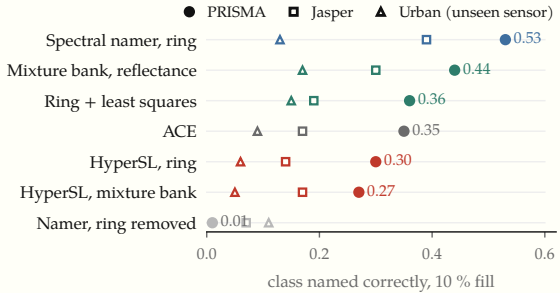
$$\mathcal{L} = \text{BCE}(\ell, \text{M}) + \text{CE}(S_{0..6}, c) + 0.5\,|\hat{a} - a|$$

| | |
|---|---|
| parameters | 347,685, of which the shared stem 17,825 |
| optimiser | AdamW $3\times10^{-4}$, stem at 0.3× |
| run | 40 epochs × 500 steps on one H100 |
| intervals | 500 resamples of 221–254 spectra per sensor |

About 34 unseen man-made spectra per sensor: differences much below 0.12 are not resolved.

# A3 HyperSL detail

PLANTED TARGETS, 10 % FILL, 8 CHOICES, TARGET'S OWN SPECTRUM REMOVED




The background estimate matters most: ring of neighbours > scene mean > nothing. Without its ring the namer collapses (0.01): its call is, in effect, "this pixel differs from its neighbours". About 40 % of targets cannot be named even when pure; that caps every method near 0.6.

PER-SCENE MIN-MAX, WORKED

Scene 1: $p_1 = (0.10, 0.30)$, $p_2 = (0.20, 0.50)$, $p_3 = (0.30, 0.40)$ give $p_2' = (0.5, 1.0)$. The same $p_2$ in scene 2 gives $(0.43, 0.75)$. One bright anomaly moves every other pixel's input. Angle $p_2, p_3$: 15.1° before, 36.9° after.

THE DECISION

| need | HyperSL | Indradhanu |
|---|---|---|
| same spectrum, same input | min-max per scene | fixed curve over wavelength |
| library spectra have an input | none | same stem |
| band lists with gaps | groups by band position | 40 nm wavelength bins |
| a pixel never rebuilds itself | rebuilds from its own bands | hides whole pixels |
| cloud in pretraining | cloud-free | cloud and haze seen |

HyperSL ran on a 149-band PRISMA axis with its default mask ratio; nothing tuned on the bench. Subtle tier suffers most: 29.7 % found against 35.0 % with no model.

# A4 Evaluation method

### PARTIAL AUC, THE HEADLINE MEASURE

$$\mathrm{pAUC}_m = \frac{1}{m}\int_0^m \mathrm{TPR}(f)\,df = \frac{1}{P}\sum_p \frac{\max(0,\, m - \mathrm{FPR}(p))}{m}$$

$m = 0.01$; $\mathrm{FPR}(p)$ is the share of background pixels scoring at or above target pixel $p$.

### PAIRED OBJECT BOOTSTRAP

- Resample the 909 objects with replacement, 2,000 draws; score both methods on each draw; report the 95 % interval of $B - A$.
- Objects, not pixels, are the independent unit; pairing removes shared object difficulty.
- It does not cover checkpoint-to-checkpoint variation, so three neighbouring checkpoints were checked.

### ALL MEASURES; EXCL. = THICK CLOUD LEFT OUT

| method | pAUC all | pAUC excl. | top 1 % all | top 1 % excl. | top 0.1 % excl. | ROC excl. |
|---|---|---|---|---|---|---|
| V2 Heavy | 0.175 | **0.189** | 422 | **449** | 16.0 % | 0.869 |
| V2.0 (FT04)* | 0.162 | 0.170 | 415 | 418 | 15.8 % | 0.852 |
| no model (PCA-RX) | 0.128 | 0.139 | 361 | 377 | 8.5 % | 0.842 |
| MNF-RX | 0.075 | 0.097 | 249 | 280 | 5.2 % | 0.753 |
| HyperSL residual | 0.083 | 0.095 | 350 | 369 | 6.7 % | 0.761 |
| HyperSL own error | 0.018 | 0.018 | 130 | 137 | 4.3 % | 0.628 |
| random | | | 93 | 93 | 1.1 % | 0.5 |

* FT04 with its own cloud and thin-cloud mask.

### WHY ONLY THICK CLOUD

Heavy's thin-cloud class fires on haze: 75.4 % of Set-4's pixels and 209 of its 230 targets. Cloud is out of scope; water is in, because vessels are targets.

# A5 Deployment

### air-gapped evaluation server

- One H200 (140 GB) with a 16-core CPU and 64 GB RAM; no network.
- The product (web API, worker, frontend, database) and the atmospheric-correction container arrive as offline archives with checksums; IT scans each image.
- Update and rollback replace containers and never touch data volumes or settings; rehearsed on a simulated copy of the server.
- Atmospheric correction runs with networking disabled, as non-root, on GPU or CPU only. DEM tiles (Copernicus 30 m South Asia, 90 m worldwide) are staged locally.

### atmospheric-correction container

| | |
|---|---|
| core | ISOFIT 4.1.5, sRTMnet emulator, 6S compiled for x86-64-v3/v4 |
| tests | 81 passed, 54 % coverage |
| size | 22 GB image, 8.98 GB compressed |

### worker cost, one PRISMA scene, 185 bands (s)

| | reconstruct | segment | score | total |
|---|---|---|---|---|
| Heavy, 2 threads | 1629 | 185 | 4 | 1819 |
| Heavy, 16 threads | 402 | 47 | 4 | 454 |
| FT04, 16 threads | 236 | 32 | | 274 |

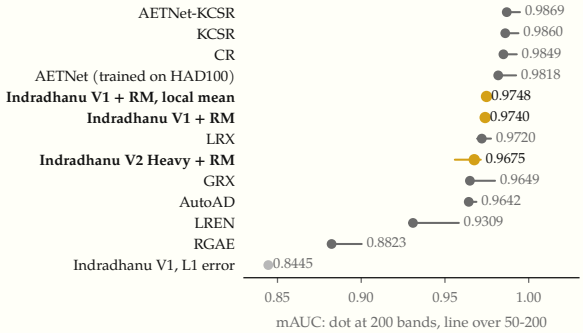
### superpixel size against accuracy, AVIRIS-NG window vs JPL

| superpixels, per-pixel step | time | land MAE | median angle |
|---|---|---|---|
| 50, analytical | 40.8 min | 0.0056 | 2.15° |
| 200, analytical | 17.8 min | 0.0056 | 2.17° |
| 200, empirical line | 11.3 min | 0.0056 | 2.17° |
| 400, empirical line | 7.5 min | 0.0057 | 2.19° |

Before the speed work: up to 75+ min for PRISMA and EnMAP and nearly 6 h for a full AVIRIS-NG scene; 12–14 GB scenes now fit the sub-2-hour window.

# A6 HAD100

PUBLIC BENCHMARK, OFFICIAL PROTOCOL, AGAINST LI ET AL. 2023




*At 200 bands residual Mahalanobis sits 0.013 (V1) and 0.019 (V2 Heavy) below the best, above GRX and the deep reconstruction baselines.*

On 64×64 crops plain RX already scores 0.96–0.98, so HAD100 ranks methods differently from our PRISMA bench.

PROTOCOL

100 crops of 64×64 from 94 AVIRIS-NG tiles (3,567 anomaly pixels, 460 objects), L1B radiance, 276 "good" bands; papers then take the first 50 / 100 / 200. Measure: mAUC, the mean per-crop ROC-AUC. Our GRX reproduces the paper's GRX to four decimals.

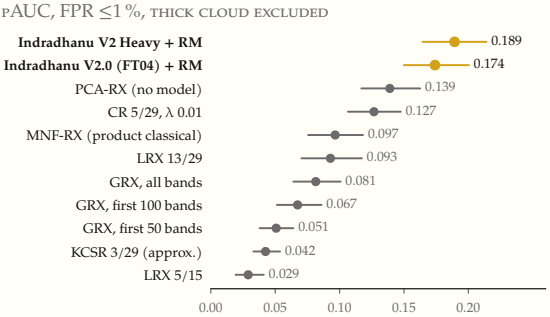
OUR RUNS, MAUC

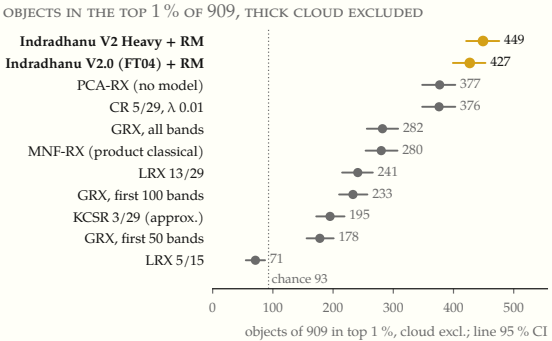
| | 50 b | 100 b | 200 b | 276 b / full | no Kakinada |
|---|---|---|---|---|---|
| GRX | 0.9799 | 0.9714 | 0.9649 | 0.9634 | 0.9611 |
| PCA30-RX | 0.9797 | 0.9603 | 0.9636 | 0.9648 | 0.9627 |
| Indradhanu V2 Heavy + RM | 0.9714 | 0.9559 | 0.9675 | 0.9703 | 0.9684 |
| local mean 5/15 | 0.9715 | 0.9565 | 0.9688 | 0.9714 | 0.9696 |
| Indradhanu V1 + RM | | | | 0.9740 | 0.9724 |
| local mean 5/15 | | | | 0.9748 | 0.9732 |

V1 reads only its 165-band grid, so one value fills every column. "No Kakinada": 94 crops, at full bands, after removing 6 from a line flown 1–2 km from a training swath.

**Reading.** V2 Heavy ties V1 (0.970 vs 0.974 at full bands; 47/52 crops, $p = 0.5$) on an airborne radiance benchmark outside its training domain (2–8 m, US/Swiss), while reading every band count natively; V1 needs its fixed 165-band grid. V2 beats GRX and PCA30-RX at 200 and 276 bands, not at 50. The leaders are local collaborative-representation detectors; AETNet trains on HAD100's own normal tiles.

# A7 HAD100 methods on our bench




*None of the HAD100 methods comes close: the best, CR, reaches 0.127 against 0.189* ($-0.063$ [$-0.090$, $-0.035$]) *and ties the no-model PCA-RX on objects* (*376 vs 377; V2 Heavy 449*).

The HAD100 ranking (KCSR > CR > LRX > GRX) does not carry over to 30 m PRISMA. Windows are HAD100's settings; CR's $\lambda$ was picked on the bench (optimistic); CR and KCSR run on a 30-dimensional projection keeping 99.97 % of the energy; KCSR approximated. AutoAD on set 1 only: 0.103 against V2 Heavy 0.262. Dot pooled over four scenes, line 95 % paired bootstrap.